

Highlights

A divide-and-conquer machine learning transition model for airfoil flows: from steady linear-lift to unsteady stall stage

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- A steady/unsteady integrated machine learning transition model is established.
- Divide-and-conquer strategy is adopted to model regions with distinct flow physics.
- Superior accuracy is achieved over the original model.

A divide-and-conquer machine learning transition model for airfoil flows: from steady linear-lift to unsteady stall stage

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ABSTRACT

In this paper, a steady/unsteady integrated machine learning transition model is established via the divide-and-conquer (DaC) strategy, which covers a wide operating range of an airfoil from linear lift to deep stall. To consider the essential distinctions across different regions of the entire flow domain, a random forest binary classifier is adopted to detect the flow field as either intermittent spot or non-intermittent spot region. Two transitional intermittency factor regressors are crafted respectively within their own regions, thus facilitating the capture of crucial phenomenological details. Three airfoil geometries with different angles of attack, Mach numbers and Reynolds numbers are selected to fully validate the newly developed DaC model. The *a posteriori* results demonstrate that the DaC model aligns well with its benchmark counterpart across the full range of angles of attack, and can overcome the limitations of Non-DaC model in the stall stage of an airfoil frequently characterized by unsteady behavior. Such divide-and-conquer-guided framework may advance the application of machine learning-based transition model to the airfoil design engineering.

1. Introduction

One of the important engineering applications of computational fluid dynamics (CFD) simulations is to predict the aerodynamic forces of an aircraft over the entire range of angle of attack (AoA), which is directly associated with the evaluation of its aerodynamic performance especially for the critical phases of takeoff and landing. Thereinto, the laminar-turbulent transition phenomenon frequently dominates the level of aerodynamic forces [1]. With the accelerating evolution of computational science and technology, numerical methods, especially the transition-integrated Reynolds-averaged Navier-Stokes (RANS) modeling strategies, have become the workhorse for transitional flows in engineering applications [2]. Different from the full-turbulence RANS model, the transition transport equation is usually introduced to formulate the transition-predictive RANS model, which constitutes the underlying logic for accurate capture of transition process. It is demonstrated that the transition-turbulence model outperforms its “fully-turbulent” counterpart over extensive engineering transitional flows such as flat-plate boundary layer [3–5], turbomachinery configurations [3–5], and high-speed straight/flared cones [6–10].

Wang and Xiao [1] comprehensively study the transition characteristics of aeronautical airfoil with the increase of AoA. It is observed that in the linear-lift stage, no apparent disparity exists between the original and transition-enabled RANS models other than the capture of laminar separation bubble (LSB) near the transition region of airfoil by the latter. As the AoA increases, the airfoil flow predicted by the transition-turbulence model shifts toward the stall stage, which is anastomotic with the experimental counterpart. Unsteady calculation is adopted to resolve the resultant nonclassical LSB [11] and turbulent separation bubble near trailing edge, which are essential to acquire simulation-experiment agreement [12]. In this scenario, transition-enabled RANS model exhibits a more consistent predictive nature with the real physical world compared with the full-turbulence RANS model. Similar conclusions are also drawn by other researches [13–15].

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Nevertheless, the employment of a transition-predictive model merely opens up the possibility of capturing transition behavior. The accuracy of the model highly depends on the comprehension of insufficiently characterized transition mechanisms (e.g., the overshoot phenomena in high-speed transitional flows [16]) and rational modeling of transition criteria. Such deficiencies are challenging to address from the perspective of traditional approaches. Recently, by leveraging the state-of-the-art machine learning (ML) algorithms, versatile and agile modeling framework can be customized to tackle numerous challenges in fluid mechanics [17, 18]. Duraisamy et al. [19] pioneer the adoption of ML techniques in the realm of transition modeling, in which a Bayesian-based field inversion and machine learning-based correction factor reconstitution (FIML) framework is proposed to improve the accuracy of traditional intermittency factor model in bypass transitional flow. Following such a practice, several research works have been conducted to modify and enhance the traditional transition models for various flows, including low-speed airfoil flows [20], flat plate/high-pressure-turbine cascade cases [21], and hypersonic flat plate/blunt cones [22]. Different from the aforementioned research works, another route for ML-enhanced transition modeling is to reconstruct the specific terms within the conventional transition model rather than to inject a spatially varying augmentation field into the standard transition equation. Inheriting from the gene expression programming (GEP) by Weatheritt and Sandberg [23] and CFD-driven framework by Zhao et al. [24], Akolekar et al. [25] develop more accurate expressions for laminar kinetic energy (LKE) production and transfer terms of the LKE transition model, thus improving the wake-mixing prediction over baseline model. Similar work can also be found in the latest study by Fang et al. [26], who adjust the laminar eddy viscosity within the production term of the LKE model.

Although the aforementioned ML-augmented transition models have various degrees of success in their own way. The existence of transition transport equation invokes more ad hoc simplifications and assumptions to close various terms as opposed to the fully turbulent RANS models [27]. Meanwhile, it consumes a higher amount of computational time and resources than its fully-turbulent counterpart. These long-standing dilemmas have motivated a more transformative paradigm shift, wherein the partial differential equation (PDE)-based transition model is substituted with the remarkable ML techniques. Recently, an artificial neural network (ANN) modeling framework has been developed for transition intermittency factor γ in Menter's shear stress transport (SST)- γ [28] and Spalart-Allmaras (SA)- γ models [29]. It is reflected that the SST- γ_{ANN} and SA- γ_{ANN} models inherit the precision of benchmark SST- γ and SA- γ models, and fall in vicinity with high fidelity (Hi-Fi) results [30, 31]. Similar studies are also conducted by other researchers [32, 33]. The engineering applicability of these models stems from the fact that their off-the-shelf baseline models demonstrate satisfactory accuracy for the flows of interest. For models below expectations, an astute and viable strategy is the augmentation-substitution modular or integrated modeling. Aiming at the practical aeronautic flows with distinct three-dimensional property, we subsequently implement the crossflow (CF) extension to augment the original γ model [28], thus deriving a CF-enhanced ANN-alternative transition model [34]. From the integrated modeling perspective, Hi-Fi simulation data are utilized by Li et al. [35] to algebraically calculate the turbulence intermittency via a neural network, which is then employed to construct an algebraic transition model.

Despite their towering empirical promise and some preliminary success, the augmentation-substitution modular and integrated modeling framework are still in their infancy regarding unsteady transition process, which is of practical relevance in a wide variety of engineering adoptions. For airfoil flow from linear-lift to stall stages as mentioned above, unsteady non-classical LSB and turbulent separation bubble appear when shifting toward the stall stage [1]. In this scenario, our previous ML-alternative transition model [30, 31] developed based on steady-state transitional airfoil flow fields is prone to collapse, thus hampering the comprehensive evaluation of airfoil aerodynamic performance. Therefore, bridging the gap between previous ML transition model for steady flow regime and that for unsteady case is the inevitable next phase.

As of now, there have been several important lessons in the community of ML-based modeling for unsteady laminar-turbulent transitional flows, which may help inform the path forward. Overall, integrating multiple time snapshots has proven effective in unsteady ML-based turbulence modeling assignment. Zhang et al. [36] develop a data-driven unsteady RANS framework to analyze the evolutionary characteristics of flow owing to cavitation-induced phase transitions. To capture the unsteady process, data with multiple time frames are fused to constitute the whole training set. Such data fusion approach is also employed in ML closure for instantaneous local eddy viscosity in particle-laden jet flows [37]. As the first effort toward unsteady FIML, Fang and He [38] propose an open-source FIML framework for spatial-temporal-accurate 45° ramp flow, in which hundreds of more flow field frames are utilized than in steady FIML.

Building upon the insights and findings of these pioneering efforts, we attempt to break the limitation of previous machine learning transition model (i.e., SST- γ_{ANN} model) that is confined to steady-state transitional airfoil flow

scenarios [30], and further establish a steady/unsteady integrated data-driven transition model spanning from linear-lift to stall stages. Here, the concept of “steady/unsteady integrated” contains three crucial aspects. The first one is that the range of steady-state, i.e., the AoA values for which the steady simulation is suitable, can be obtained independently through machine learning transition model without reliance on baseline model. The same also goes for the unsteady-state cases. The second one is that the results from steady and unsteady simulations must remain consistent in steady-state regimes. The third one is that steady method is invalid for unsteady cases, necessitating reliance on unsteady simulations. These serve as the fundamental requirements for engineering-practical model of transitional airfoil flow across the entire range of $AoAs$.

The rest of this paper is organized as follows. First of all, we review in Sec. 2 the previous machine learning steady-state transition modeling framework and the multi-time fusion strategy employed in present model, together with its catastrophically wrong predictions and underlying *a priori/a posteriori* reasons. Such a dilemma is expected to be overcome via the divide-and-conquer (DaC) technique in this study, we then introduce in Sec. 3 the DaC-based data-driven turbulence/transition modeling strategy, the framework of established DaC model, and each component within the DaC model. Furthermore, three airfoils with different flow regimes are selected in Sec. 4 to systematically investigate the performance of Non-DaC and DaC models with reference to the experimental data. Finally, the main findings and remaining limitations are highlighted in Sec. 5.

2. The non-divide-and-conquer model

2.1. Previous steady SST- γ_{ANN} model

To consider the transition effect, Menter et al. [28] propose a local correlation-based model equation of intermittency factor γ in the form

$$\frac{\partial(\rho\gamma)}{\partial t} + \frac{\partial(\rho u_j \gamma)}{\partial x_j} = \frac{\partial}{\partial x_j} \left[\left(\mu_L + \frac{\mu_T}{\sigma_\gamma} \right) \frac{\partial \gamma}{\partial x_j} \right] + P_\gamma - E_\gamma, \quad (1)$$

in which the intermittency factor is presumed to be zero in laminar boundary layer upstream of the transition region and is unity in the turbulent boundary layer region. μ_L and μ_T denote the molecular and turbulent viscosity, respectively. P_γ and E_γ represent the production and destruction terms in the γ transport equation which respectively read

$$P_\gamma = F_{\text{length}} \rho S \gamma (1 - \gamma) F_{\text{onset}}, \quad (2)$$

and

$$E_\gamma = C_{a2} \rho \Omega \gamma F_{\text{turb}} (C_{e2} \gamma - 1). \quad (3)$$

Incorporation between the fully-turbulent $k-\omega$ SST model [39] and above γ model is achieved by modifying the production and destruction terms of the turbulent kinetic energy (TKE) equation for the original SST turbulence model. Meanwhile, an extra production term P_k^{lim} is introduced for the TKE equation. The resultant transport equations of TKE (k) and specific turbulence dissipation rate (ω) are respectively written as

$$\frac{\partial(\rho k)}{\partial t} + \frac{\partial(\rho u_j k)}{\partial x_j} = \frac{\partial}{\partial x_j} \left[\left(\mu_L + \sigma_k \mu_T \right) \frac{\partial k}{\partial x_j} \right] + \gamma \tau_{ij}^F S_{ij} + P_k^{\text{lim}} - \max(\gamma, 0.1) \beta^* \rho \omega k, \quad (4)$$

and

$$\frac{\partial(\rho \omega)}{\partial t} + \frac{\partial(\rho u_j \omega)}{\partial x_j} = \frac{\partial}{\partial x_j} \left[\left(\mu_L + \sigma_\omega \mu_T \right) \frac{\partial \omega}{\partial x_j} \right] + \frac{C_{\omega\rho}}{\mu_T} \tau_{ij}^F S_{ij} - \beta \rho \omega^2 + 2(1 - f_1) \frac{\rho \sigma_{\omega 2}}{\omega} \frac{\partial k}{\partial x_j} \frac{\partial \omega}{\partial x_j}, \quad (5)$$

which are solved to calculate the turbulent viscosity μ_T , thus closing Reynolds stress tensor τ_{ij}^F in RANS equation. For brevity and simplicity, readers are referred to the original literature [28, 39] for more details about the functions (S , Ω , F_{onset} , F_{turb} , P_k^{lim} , S_{ij} , and f_1), constants (σ_γ , F_{length} , C_{a2} , C_{e2} , σ_k , β^* , σ_ω , C_ω , β , and $\sigma_{\omega 2}$), and formulations of the SST/SST- γ models.

Compared with the full-turbulence SST model, implementation of γ leads to a more consistent generation of TKE with the true physical process, thus contributing to a more accurate prediction of turbulent viscosity μ_T and Reynolds stress τ_{ij}^F . Considering the imperative role and further deficiencies of additional transport equation of γ as mentioned in Sec. 1, we have resorted to bypass the solving of PDE-constrained γ and reconstruct the functional form of it using

ANN algorithm [30]. To be specific, ANN transition model is built to map the relation between RANS mean flow variables $\mathbf{q}(\mathbf{x})$ and space-dependent intermittency factor $\gamma(\mathbf{x})$ using the data obtained from traditional SST- γ model, which is then fully substituted for traditional γ equation during CFD iteration. Despite its remarkable accuracy and generalizability for transitional airfoil flows, unsteady flow regimes still present considerable difficulties.

2.2. Unsteady extension to steady SST- γ_{ANN} model

With particular emphasis on the full description of airfoil flows from linear-lift to stall stages, unsteady extension is implemented via the well-recognized multi-time fusion strategy [36–38]. To generate the training and testing datasets, an in-house code is utilized, which integrates the two-dimensional RANS and SST- γ equations. Inviscid fluxes are discretized using the second-order central scheme with artificial diffusivities [40], and the viscous fluxes are discretized using traditional second-order central finite difference scheme. The temporal advancement is achieved by using the lower-upper symmetric Gauss-Seidel (LU-SGS) method with/without sub-iterations for unsteady/steady simulations, respectively. Six instantaneous flow fields past the NACA0015 airfoil at $AoA = 15.00^\circ$, $Ma = 0.378$, and $Re = 1.95 \times 10^6$ are adopted as the training set, and two counterparts of them are reserved for validation. The new unsteady-extended SST- γ_{ANN} model is constructed in a similar way to previous steady-state model [30]. For brevity, the details of model construction and its hyperparameter are summarized in Table 1. Thereinto, λ_1 , λ_2 , and λ_3 correspond to the coefficients of loss function in the form

$$\frac{\lambda_1}{N} \sum_{i=1}^N (\gamma_i - \hat{\gamma}_i)^2 + \frac{\lambda_2}{N} \sum_{i=1}^N \text{ReLU}(-\hat{\gamma}_i) + \frac{\lambda_3}{N} \sum_{i=1}^N \text{ReLU}(\hat{\gamma}_i - 1.5), \quad (6)$$

in which γ_i and $\hat{\gamma}_i$ represent the true and predicted outputs of the i th sample, respectively, and the last two terms are designed to constrain the model predictions within range of 0 to 1.5. Through the far-reaching permutation feature importance (PFI) method [41] (not shown here for brevity), sixteen input features are extracted from the candidate inputs which are all presented in Appendix.

Table 1
Summary of the details for present ANN model.

Details	Description
Input features	$q_1, q_2, f_1, f_2, f_3, f_5, f_6, f_7, f_9, f_{11}, f_{12}, f_{16}, p_2, p_3, p_5, p_7$
Model architecture	(16, 128, 64, 1)
λ_1	1.0
λ_2	1.0
λ_3	1.0
Optimizer	Adam
Learning rate	Decreasing dynamically
Activation function	LeakyReLU with slope coefficient being 0.05

Figure 1 tracks the behavior of loss function over the course of training epochs. To facilitate an intuitive insight into the performance of this modeling strategy in unsteady flow regimes, we also present result trained with our previous steady datasets [30] while keeping other arguments stationary (labeled as “Steady Data” in Fig. 1). Obviously, the model derived from unsteady flow fields lags far behind that obtained through steady-state data. Therefore, the well-validated multi-time fusion strategy appears to fail in current unsteady transition modeling issue from the *a priori* perspective. Depicted in Fig. 2 are variations of the lift coefficient C_L and drag coefficient C_D with respect to AoA from *a posteriori* test cases of NACA0015 airfoil at $Ma = 0.291$ and $Re = 1.95 \times 10^6$. The legend with “Non-DaC” refers to the ANN-partially-substituted SST- γ model in this section. In the linear-lift stage ($AoA \leq 12^\circ$), all three models exhibit favorable agreement with the experimental measurement by Piziali [42], except for the overestimation of drag coefficient C_D using fully-turbulent SST model. In this scenario, all flow fields are basically dominated by steady behavior and thus can be derived by the steady-state time-marching LU-SGS method. Moreover, the results from traditional SST- γ and ANN-substituted SST- γ_{ML} models overlap each other, which also holds for distributions of mean pressure coefficient $C_p = 2(p - p_\infty)/(\rho_\infty u_\infty^2)$ and mean skin friction coefficient $C_f = 2\tau_w/(\rho_\infty u_\infty^2)$ (see Figs. 3 (a) and (b)). These consistencies signify the achievement of current ANN transition model in steady flow regimes similar to previous study [30]. However, as the AoA increases beyond 12° , apparent disparity exists between

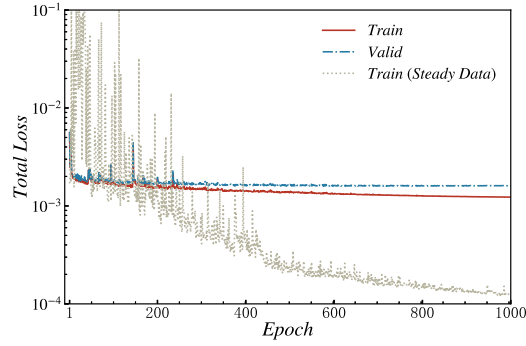


Fig. 1. Evolutions of the loss function for training and validation datasets.

the SST- γ and SST- γ_{ML} models, accompanied by larger deviation of the SST- γ_{ML} model from experiment. Similar phenomena also appear for C_p and C_f in Figs. 3 (c) and (d), in which a double suction peak emerges in C_p from SST- γ model. In this scenario, LU-SGS strategy with inner iterations is the only enabler in SST- γ model owing to the arisen unsteadiness. By contrast, steady characteristics mistakenly persist in SST- γ_{ML} model (also SST model), rendering steady LU-SGS method valid over the entire range of AoAs. Therefore, it is evident that the unsteady extension (i.e., universally-acknowledged multi-time fusion strategy) remains problematic for the current transition modeling issue presented in this paper.

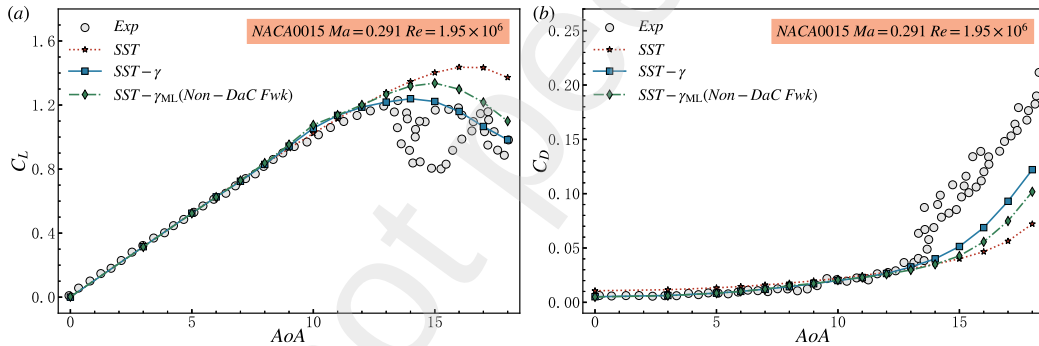


Fig. 2. Integral forces for different angles of attack: (a) lift coefficient and (b) drag coefficient from present numerical simulations and previous experimental measurements [42].

2.3. Underlying reasons for model performance

To probe the underlying reasons for deteriorative performance revealed by the above SST- γ_{ML} model, several inspections are made from both the offline and online perspectives. For the former, it is qualitatively unveiled that the ML prediction achieves excellent agreement with that of benchmark model (not shown here for brevity), except for the transition region of airfoil as detailed in Fig. 4. Compared with the baseline model, ML model fails to capture the phenomenological detail of irregular and fragmented spots, which frequently fluctuate between high and low values and are predominantly concentrated in two regions labeled as “Reg 1” and “Reg 2” in Fig. 4. Such phenomena are similar to the turbulent spots found in transition-turbulence studies using Hi-Fi methods [43]. To describe them scrupulously, such regions are conceptually termed as “intermittent spot region (ISR)” hereafter. With this in mind, quantitative investigation is conducted by training another ANN model without ISR data. The evolution of loss function is described in Fig. 5 with “without ISR” label. The other three curves keep the same as Fig. 1. As can be seen, after removing the ISR data, the degree of convergence is improved to a satisfactory level similar to that trained by steady data. Therefore, the existence of ISR is the overarching or even only cause of high training and validation loss. In other words, the present modeling strategy is presumed to be valid for the whole region except for ISR.

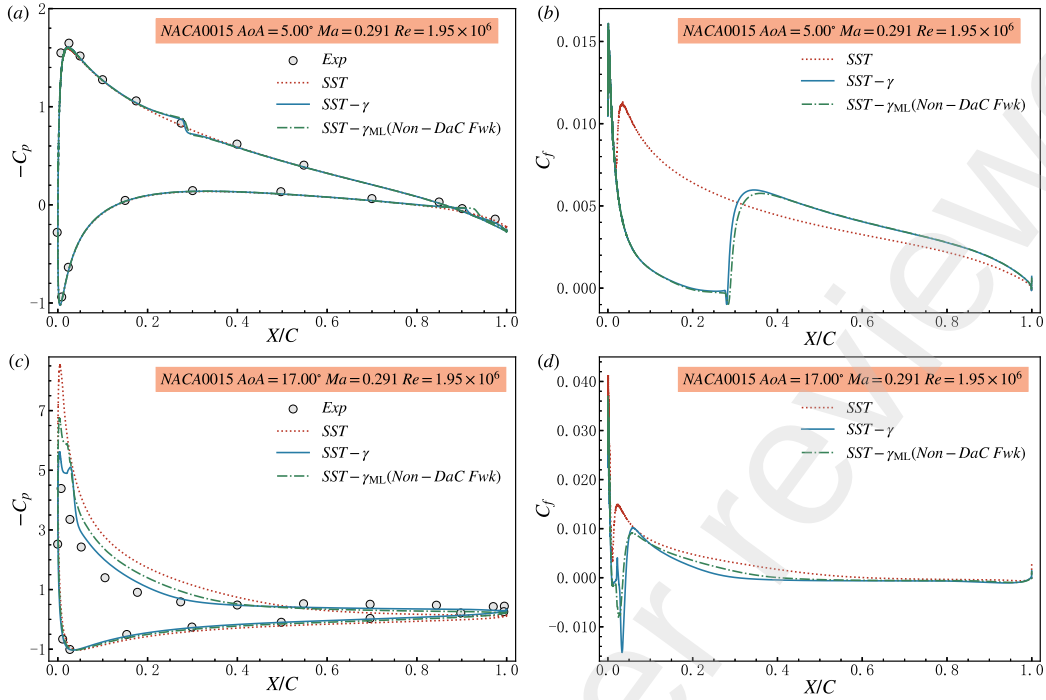


Fig. 3. Distributions of mean pressure coefficient C_p ((a) and (c)) and mean skin friction coefficient C_f ((b) and (d)) (suction side only) from present numerical simulations. The experimental data from Piziali [42] are plotted for comparison, and the corresponding flow regime is labeled in each panel for clarity.

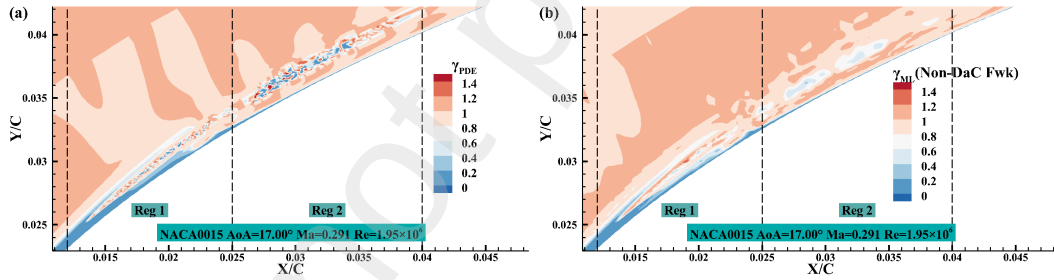


Fig. 4. Zoom-in contours of instantaneous intermittency factor γ around the transition region of NACA0015 airfoil at $AoA = 17.00^\circ$, $Ma = 0.291$, and $Re = 1.95 \times 10^6$: (a) Traditional model and (b) ML model.

Despite the significant inroads into the reason of deteriorative *a priori* performance, several imperative issues still remain to be tackled. Does the unresolved two ISRs contribute to the failure of *a posteriori* prediction over unsteady flow regimes? If so, is there a difference in the importance of these two ISRs? If so, which one is more important? To answer these questions, we design a hybrid transition modeling framework. To be specific, the intermittency factor γ of the two ISRs are alternately (or simultaneously) governed by traditional transition model, whereas the ML model retains to predict the remaining region during the CFD iteration. As exhibited in Fig. 6, there is a sharp contrast in the significance of the two ISRs for *a posteriori* predictions. Provided that γ in upstream ISR is predicted by traditional transition model, corresponding results achieve an excellent agreement with the benchmark SST- γ model, while the downstream ISR proves to be of negligible importance. Consequently, the declining consistency between SST- γ and SST- γ_{ML} models in unsteady flow scenario originates from the disability of ML transition model to capture the upstream ISR. This encouraging finding will inform the further design of model framework.

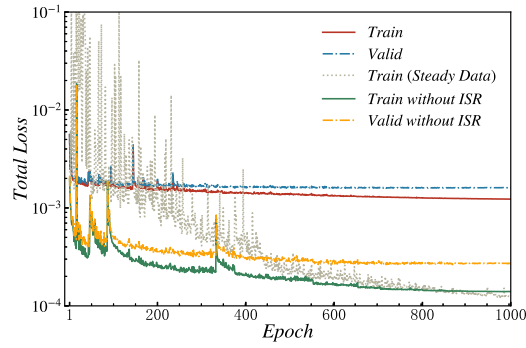


Fig. 5. Evolutions of the loss function for training and validation datasets: whole region versus that without intermittency spot region.

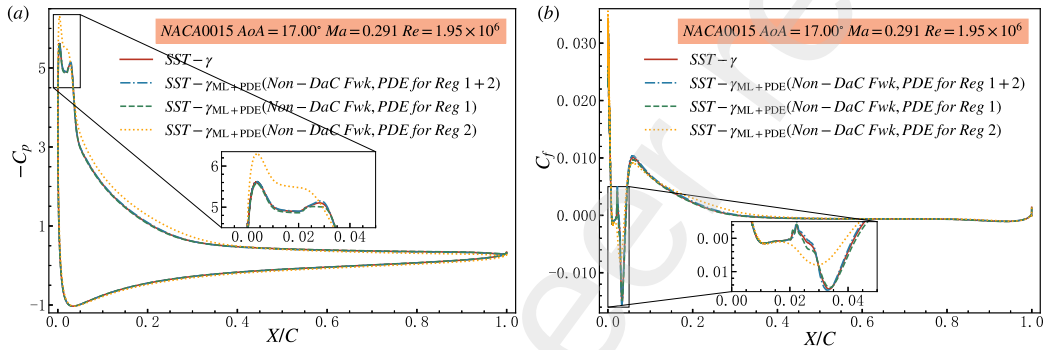


Fig. 6. Distributions of (a) mean pressure coefficient C_p and (b) mean skin friction coefficient C_f (suction side only) from SST- γ and SST- γ_{ML+PDE} models, in which the subscript “ML+PDE” indicates that the value of γ is selectively determined by ML or PDE approach. “Reg1” and “Reg2” refer to the two clustered regions of intermittency spot in figure 4 (a).

3. The divide-and-conquer model

3.1. Adoption of the divide-and-conquer strategy and literature review

As reflected in Fig. 4(a), γ within ISR is characterized by pronounced multi-scale properties, differing from that in other regions and even the input features within the same spatiotemporal domain. To further investigate the nature of γ in the upstream ISR (abbreviated as ISR henceforth), temporal characteristics of γ in ISR are depicted in Fig. 7. As can be seen, γ undergoes an exceptionally high frequency oscillations (Fig. 7(a)) and the PDF distributions demonstrate no significant patterns (Fig. 7(b)). Such spatiotemporal properties present completely different flow physics from other regions. In addition, the spatial terms of γ equation (see Eq. (1)) are calculated as shown in Fig. 8. It is apparent that the emerging unsteadiness in high AoA regimes stems exclusively from the two ISRs. In this scenario, γ in ISR is also shaped by its historical evolution, not limited to the spatial input features like non-intermittent spot region (Non-ISR).

This challenge cannot be easily overcome through a unified or even brute-force ML model, especially in the highly-parallel and localized environment of CFD. Therefore, dividing the whole-domain modeling task into several subtasks emerges as a natural strategy. Specifically, each region is assigned to its own ML model, which is optimized by its region-specific data, thus contributing to the prediction accuracy improvement of the entire domain. Such divide-and-conquer (DaC) strategy enables each model to concentrate on its specific region, where the flow physics tends to be more straightforward and simpler, resembling the mixture of experts (MoE) architecture in DeepSeek [44].

To our knowledge, the adoption of DaC ideology is not rare in turbulence modeling. The earliest and most representative work can be traced back to the detached-eddy simulation (DES) model proposed by Spalart [45]. In the community of ML-based turbulence modeling, there has been ongoing effort to develop partitioned frameworks in which their zonal scalar indicators are generally straightforward and feasible to implement in CFD iteration [24, 46–48]. However, in some flow scenarios, the partitioning criteria are not well-defined, necessitating the use of additional ML

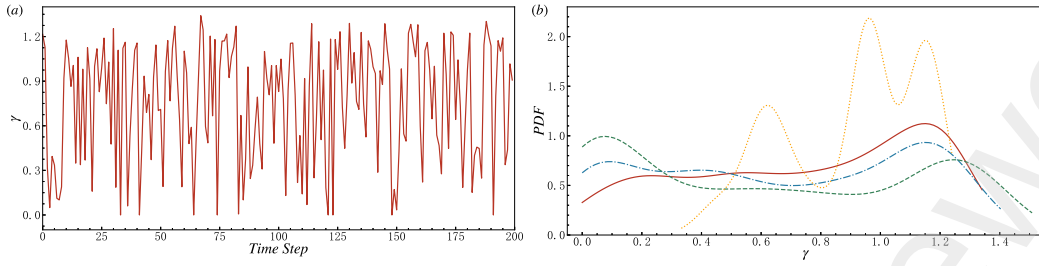


Fig. 7. Temporal characteristics of γ in ISR of NACA0015 airfoil at $AoA = 17.00^\circ$, $Ma = 0.291$, and $Re = 1.95 \times 10^6$: (a) temporal evolution (200 CFD iterations are extracted) at a single point and (b) probability distribution function (PDF) at several representative points.

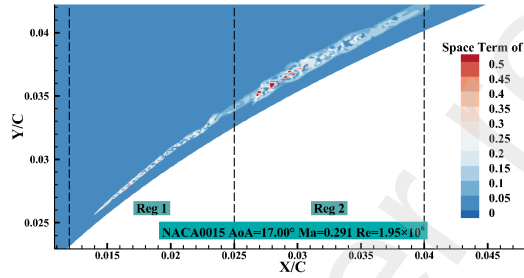


Fig. 8. Zoom-in contours of instantaneous spatial terms of γ equation (see Eq. (1)) of NACA0015 at $AoA = 17.00^\circ$, $Ma = 0.291$, and $Re = 1.95 \times 10^6$ in accordance with Fig. 4. The maximum counterpart is used for normalization.

classifiers. Ling and Kurzwski [49] are among the first to explore this approach, in which a ML classifier is proposed to provide a zonal factor f_{ML} to activate the higher-order quadratic eddy viscosity model (QEVm) in addition to linear eddy viscosity model (LEVm) in the form

$$\left(\overline{u'_i u'_j}\right)_{ZONAL} = (1 - f_{ML}) \left(\overline{u'_i u'_j}\right)_{LEVm} + f_{ML} \left(\overline{u'_i u'_j}\right)_{QEVm}. \quad (7)$$

Obviously, their zonal implementation is designed to strengthen the stability and convergence properties rather than to enhance the prediction precision. Considering the patchy correction field $\beta(x, y)$ of FIML, decision trees algorithm is employed to bin the patches into a series of particular coefficient value [50]. Despite their preliminary success, significant challenge still remains for DaC approach to exhibit its superiority over Non-DaC methods in practical engineering flows. In response to this crux, we expect the DaC method to effectively address the dilemma outlined in Sec. 2.2-2.3, thereby driving transformative progress in the development of steady/unsteady integrated data-driven transition model. On the whole, a ML classifier is developed to distinguish ISR grid points (Sec. 3.2), and two regressors are formulated to predict γ in ISR and Non-ISR, respectively (Sec. 3.3-3.4), which are sketched in Fig. 9.

3.2. IFISR classifier

Accurately labeling each grid point of entire domain is identified as the priority. Unlike the majority of previous partitioning work, separating the flow field into ISR and Non-ISR regions is not a task that can be accomplished simply through physical or *a priori* knowledge. Multiple physical quantities are explored to achieve a precise portrayal of the upstream ISR. It is revealed that F_{onset} , the model function in the production term of γ equation, behaves as a distinctly conical profile at the interface between ISR and Non-ISR (see Fig. 10). This characteristic serves as the foundation for designing ML classifier of ISR. To be specific, each grid point in the flow field is assigned a label (referred to as IFISR henceforth) of 0 (Non-ISR) or 1 (ISR) based on the value of F_{onset} . Then, a random forest (RF) classifier [51] is established to map the relation between selected input features and IFISR.

Consolidated in Table 2 are details of the model construction and its hyperparameters. The training set consists of four instantaneous flow fields with different airfoil geometries and flow regimes, while four additional counterparts under the same conditions are retained for validation. Notably, steady flow cases are incorporated into the dataset

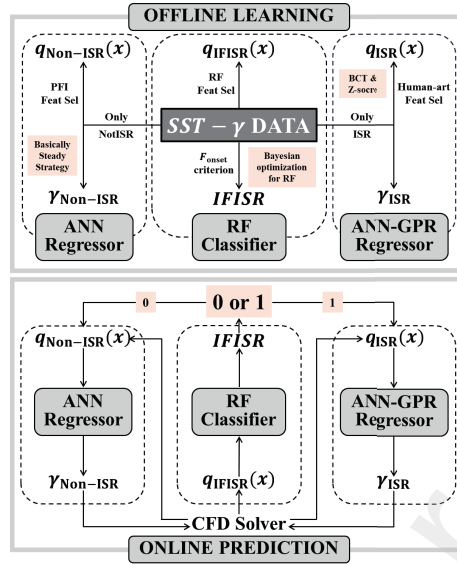


Fig. 9. The overall schematic diagram of divide-and-conquer machine learning transition model.

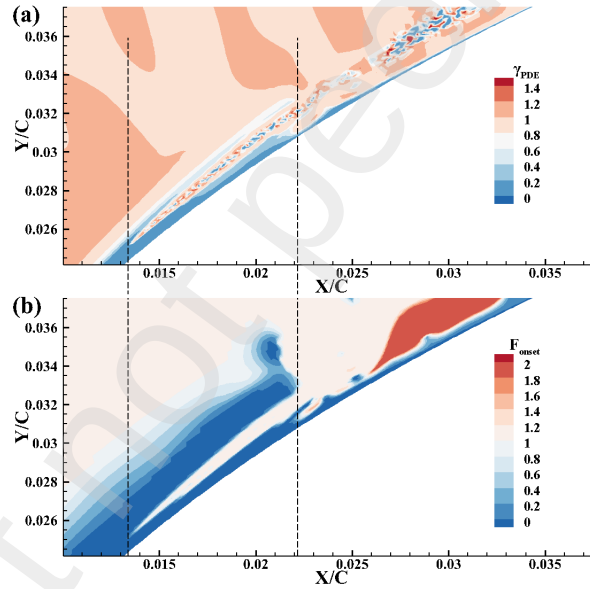


Fig. 10. Zoom-in contours of instantaneous variables in SST- γ model near the ISR of NACA0015 at $AoA = 17.00^\circ$, $Ma = 0.291$, and $Re = 1.95 \times 10^6$ in accordance with Fig. 4: (a) Intermittency factor γ and (b) model function F_{onset} .

to facilitate a steady/unsteady integrated classifier. A more concise combination of inputs is derived through the relative importance of candidate input features (listed in the Appendix) using RF model (see Fig. 11). To optimize the hyperparameters of RF classifier, Bayesian optimization process [52, 53] is employed with its objective function being the F1-score of validation data. As shown in Fig. 12, an iteration of 60 generates the best model parameters on the validation set.

Figure 13 shows the IFISR field derived from the RF classifier. As is clearly seen, the conical region is effectively extracted from the intricate flow field, thus enabling accurate detection of IFISR. This lays a solid foundation for the subsequent division of labor among the individual regression tasks.

Table 2

Summary of the RF classifier details.

Details	Description
Training set	NACA0015, $AoA = 15.00$, $Ma = 0.378$, $Re = 1.95 \times 10^6$ NACA0015, $AoA = 9.58$, $Ma = 0.265$, $Re = 1.95 \times 10^6$ NACA2414, $AoA = 15.99$, $Ma = 0.309$, $Re = 1.32 \times 10^6$ NACA2414, $AoA = 16.16$, $Ma = 0.216$, $Re = 2.67 \times 10^6$
Input features	$q_3, f_2, f_3, f_4, f_7, f_9, f_{12}, f_{13}, f_{14}, f_{15}, p_3, p_4, p_7, p_8, p_9, p_{10}, p_{11}$
n_estimators	531
max_depth	22
min_samples_split	17
min_samples_leaf	1

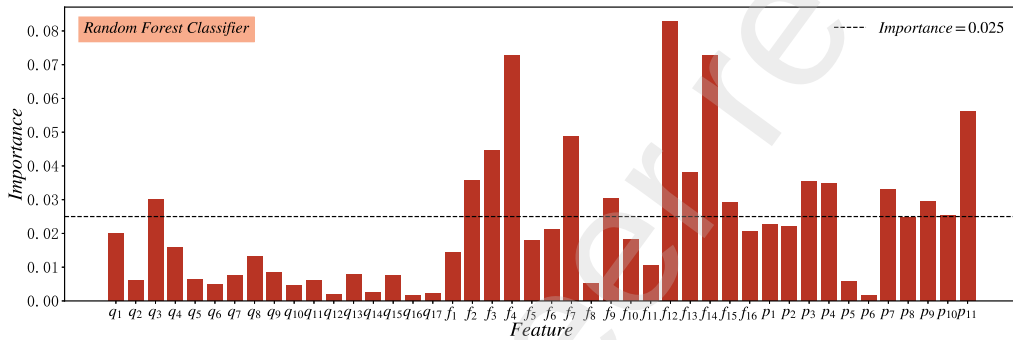


Fig. 11. Relative importance of candidate input features obtained using RF model. A threshold of 0.025 is set to compact the inputs.

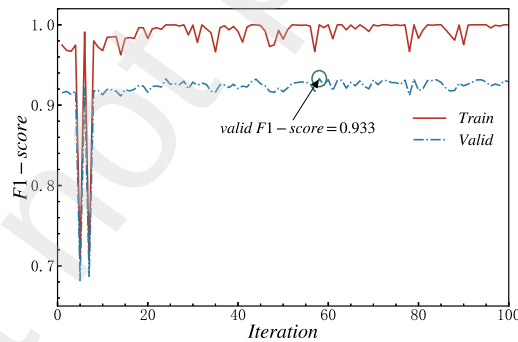


Fig. 12. Bayesian optimization process for training and validation datasets.

3.3. Non-ISR regressor

As highlighted in Sec. 2.3, the unsteady extension framework in Sec. 2.2 can be directly adopted for the current Non-ISR regressor. It should be noted, however, that there is no rationale to incorporate ISR data into the Non-ISR regressor since the Non-ISR regressor is not responsible for γ in ISR. In reality, such approach is likely to hinder the performance of Non-ISR model as we have discussed in Fig. 5.

3.4. ISR regressor

Taking guidance from the findings in Sec. 2.3, appropriate modeling of the spatiotemporal properties of γ in ISR is the key component of its unsteady extension. In view of the multi-scale and high-frequency oscillation characteristic of γ as reflected in Fig. 4(a), the Box-Cox transformation (BCT) [54] is adopted to shift raw feature x into the BCT

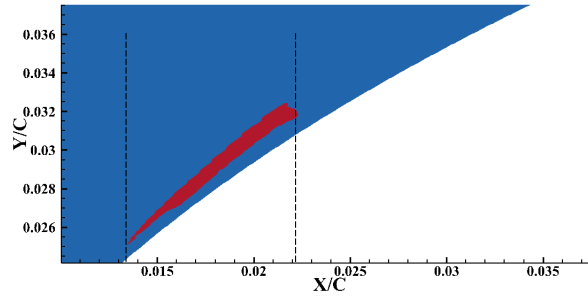


Fig. 13. Zoom-in contours of instantaneous IFISR field of NACA0015 at $AoA = 17.00^\circ$, $Ma = 0.291$, and $Re = 1.95 \times 10^6$ in accordance with Fig. 4. The two dashed lines align with those in Fig. 10.

feature $\mathcal{F}(x)$:

$$\mathcal{F}(x) = \begin{cases} \frac{x^\lambda - 1}{\lambda}, & \lambda \neq 0, \\ \log x, & \lambda = 0, \end{cases} \quad (8)$$

where λ is a hyper-parameter. In practice, x is first converted to its absolute value before undergoing the BCT, given that the input variables in this paper are not always non-negative. As depicted in Fig. 14, BCT exerts an amplifying effect on small x , mapping the low-order quantity 10^{-k} ($k \geq 2$) to $\mathcal{O}(1)$ with appropriate λ [55], and simultaneously eliminates the singularity encountered in the widely-used log transformation.

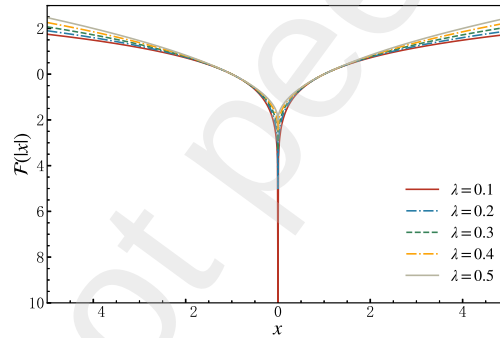


Fig. 14. Profiles of BCT with typical λ .

Depicted in Fig. 15(b) and (c) are one-dimensional distributions of two representative raw features and their BCT counterparts (with $\lambda = 0.1$) in the ISR region, where the horizontal axis denotes the spatial index. The distribution of γ is also given in Fig. 15(a) for comparison. High values in raw features are substantially compressed by BCT, whereas the small values with high-frequency oscillations are sketched to a broader representation space. This partially restores the high-frequency oscillatory structures of the modeling target of γ , making the overall distribution more aligned with γ . In addition, BCT eliminates the magnitude discrepancy across different inputs, scaling the original ranges of q_{16} and f_9 to the same order of magnitude (i.e., $\mathcal{O}(1)$), which is conducive to model training.

Different from numerous ML-based turbulence or transition modeling works, present γ in ISR is shaped not only by spatial input features but also by its own historical values as elaborated in Sec. 3.1. Several ML algorithms with spatiotemporal prediction capability (e.g., recurrent neural network and long short-term memory) have been tested, but the results fall short of expectations. Furthermore, these methods significantly increase the model complexity and place higher demands on computational and storage capacity. These drawbacks, to some extent, conflict with our original intent of the modeling framework. Considering the stochastic nature of γ in ISR, we opt for the probabilistic Gaussian process regressor (GPR) [56] for its built-in capability to model uncertainty, other than deterministic ML models of ANN and RF. To be specific, an ANN is utilized as the extractor of high-dimensional input features to project them into a more meaningful feature space, which is then processed by GPR. This ANN-GPR architecture effectively combines the representational capability of ANN with the non-parametric flexibility of GPR [57]. A series of inputs are selected

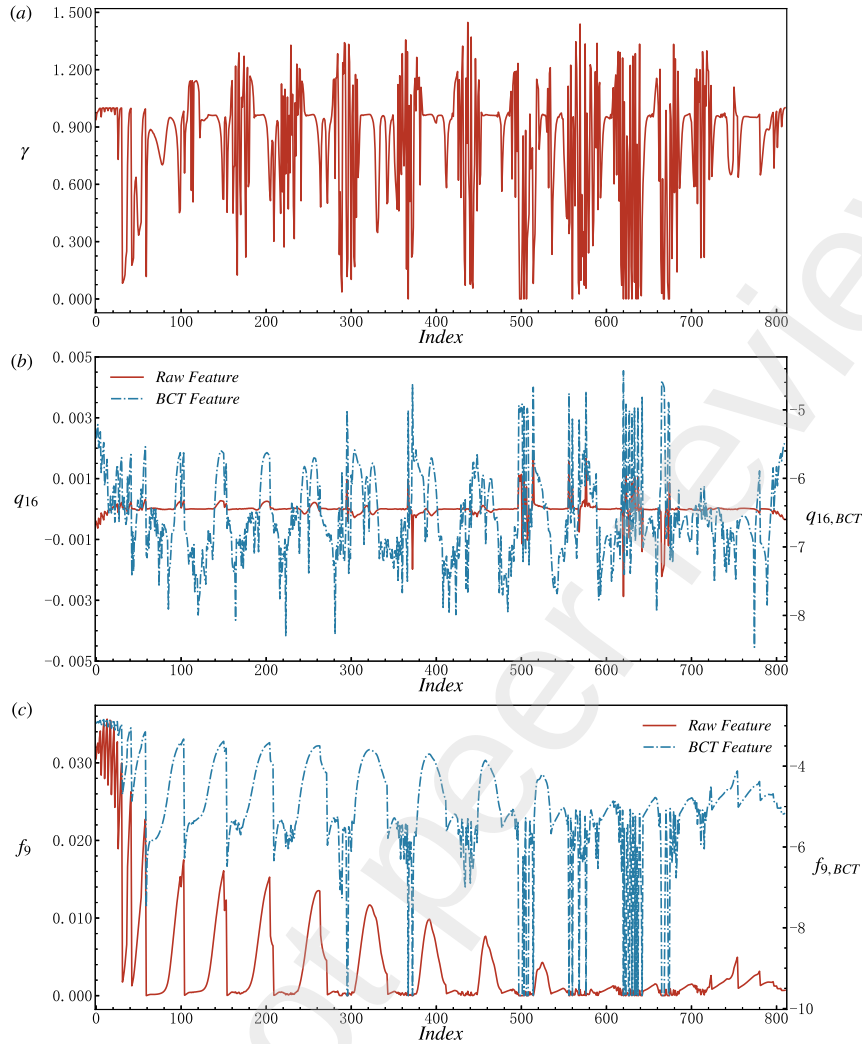


Fig. 15. One-dimensional spatial distributions of (a) intermittency factor γ , (b) q_{16} , and (c) f_9 in ISR of NACA0015 at $AoA = 17.00^\circ$, $Ma = 0.291$, and $Re = 1.95 \times 10^6$ in accordance with Fig. 4.

from the predefined input library to capture the highly oscillatory characteristics of γ as summarized in Table 3. Z-score normalization is implemented after the BCT. Other details regarding the model construction and its hyperparameters are provided in Table 3. Depicted in Fig. 16 are representative prediction results for NACA0015 airfoil at different $AoAs$. Low-frequency oscillations are observed in steady scenario at $AoA = 5^\circ$, where the ANN-GPR prediction aligns closely with the SST- γ model. For unsteady regime at $AoA = 17^\circ$, the ANN-GPR model adequately reproduces the high-frequency oscillatory patterns that are intensively emphasized in present paper with a narrow confidence interval. To compensate the omission of temporal effect of γ , upper and lower bounds of confidence interval are randomly leveraged as the practical value of γ . By doing so, the desired randomness is injected into the γ field, while maintaining its interpretability and credibility.

Table 3

Summary of ANN-GPR regressor details.

Details	Description
Training set	Same with Table 2
ANN model architecture	(16, 128, 8, 2)
Input features	$q_3, q_7, q_8, q_9, q_{11}, q_{13}, q_{14}, q_{15}, q_{16}, q_{17}, f_6, f_7, f_9, p_1, p_3, p_9$
ANN activation function	LeakyReLU with slope coefficient being 0.05
Optimizer	Adam
Mean function	Constant mean function
Kernel function	Radial basis function (RBF) Kernel
Likelihood function	Gaussian likelihood function

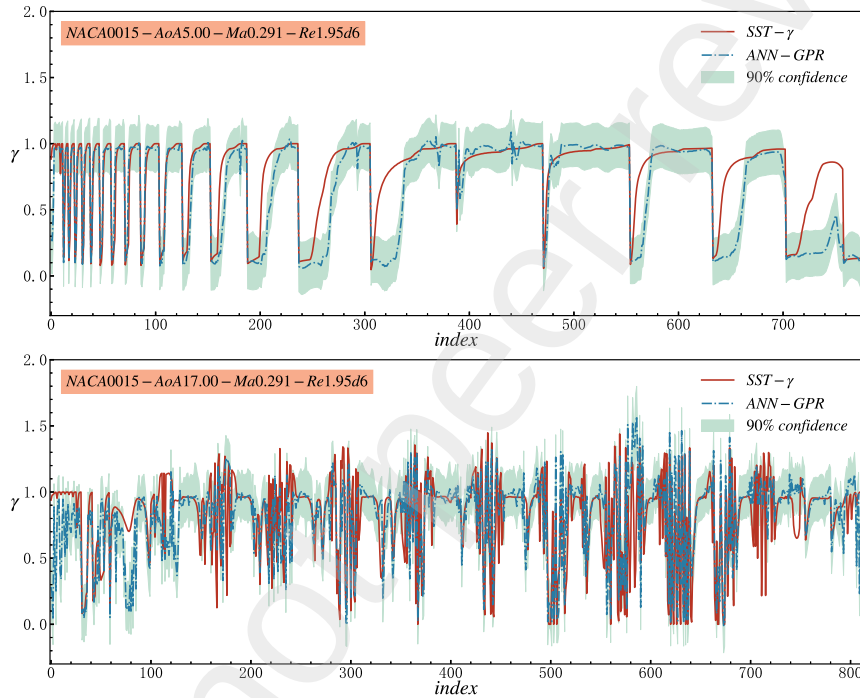


Fig. 16. One-dimensional spatial distributions of γ in ISR of NACA0015 at (a) $AoA = 5.00^\circ$ and (b) $AoA = 17.00^\circ$ given by SST- γ and ANN-GPR models in accordance with Fig. 4. The 90% confidence interval is also displayed for comparison. The corresponding flow parameters are labeled in each panel.

4. Results

4.1. Generalization about AoA and Mach number

Considering the availability of experimental data [42], the NACA0015 airfoil at the Mach and Reynolds numbers of 0.291 and 1.95×10^6 is selected to validate the newly established DaC SST- γ_{ML} model. Shown in Figs. 17(a) and (b) are the variations of lift coefficient C_L and drag coefficient C_D on AoA . Compared with the Non-DaC model in Sec. 2.2, the present DaC model addresses the limitations of the latter in unsteady prediction, and reaches a high consensus with the benchmark SST- γ model, falling in vicinity of the experimental results. Furthermore, owing to the sensitivity of IFISR classifier to AoA , the ISR regressor remains inactive in steady regimes (i.e., $AoA \leq 12^\circ$), enabling the predictions to naturally retrieve the Non-DaC model. For unsteady circumstance with $AoA > 12^\circ$, however, the accurate identification of ISR and proper modeling of γ prevent the DaC model from being solved with a steady time-marching method, resembling its baseline SST- γ model, which aligns with the “steady/unsteady integrated” requirements as underscored in Sec. 1.

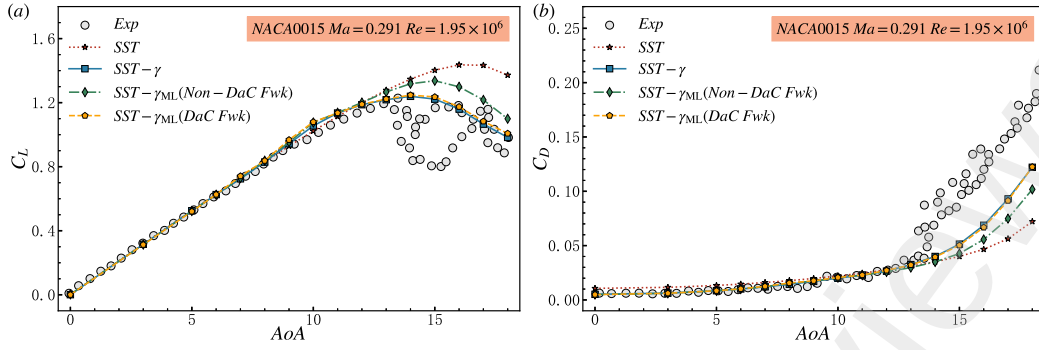


Fig. 17. Variations of integral forces with respect to angle of attack for NACA0015 airfoil at $Ma = 0.291$ and $Re = 1.95 \times 10^6$ obtained using different models: (a) lift coefficient and (b) drag coefficient. The experimental data [42] are also included for comparison.

To explore more detailed differences behind aerodynamic integral forces, the distributions of mean pressure and skin friction coefficients are calculated at six typical AoAs as shown in Fig. 18 and 19. Zoom-in streamlines around the transition region of the airfoil are also depicted accordingly in Fig. 20 to elucidate the observed discrepancies. For steady cases with $AoA \leq 12^\circ$, a unique suction peak is observed in C_p by all RANS models, and an additional small plateau region is predicted by transition-predictive RANS models, which indicates the existence of a laminar separation bubble (LSB). Subjected to an intense adverse pressure gradient, the laminar boundary layer separates, forming an LSB, and subsequently transitions to the attached turbulent boundary layer. The appearance of LSB can also be recognized by the negative region of C_f and streamlines in Fig. 19 and 20, respectively. Obviously, for steady cases, the performances of both Non-DaC and DaC models are consistent with that of the baseline SST- γ model, corresponding to the overlap of aerodynamic integral forces predicted by them.

For unsteady scenario, however, qualitative transformation can be observed for the performances of both SST- γ and DaC models (see Figs. 20(b) and (f)). The amplitude of Reynolds stress is progressively intensified, forcing the single vortex in steady LSB to break into the unsteady double-vortex structure, and thereby forming a nonclassical LSB with enhanced stability [11]. Besides, the positions and sizes of the primary and secondary vortices predicted by the two models are also close to each other. By contrast, the Non-DaC model still predicts the steady LSB with a single vortex, indicating a continuation to the steady-state feature (see Fig. 20(d)). Consequently, consistent patterns of C_p and C_f are given by the Non-DaC model across the entire range of AoAs (similar to the SST model), which lead to the deviations of C_L and C_D in unsteady flow regimes.

Taking a closer look at the double-vortex structure given by the SST- γ and DaC models, the secondary vortex is found to be surrounded by its primary vortex with the two vortices rotating in opposite directions: clockwise for the primary vortex and counterclockwise for the secondary vortex. This structural feature exerts a significant impact on the distribution of C_p . As can be seen in Figs. 18(c)-(f), after a narrow plateau region (corresponding to the primary vortex), the pressure coefficient C_p rises again to form a secondary suction peak due to the subsequent secondary separation bubble characterized by favorable pressure gradient. Therefore, as the flow converts from steady to unsteady regimes (i.e., from steady LSB to nonclassical LSB), the profile of C_p shifts from a single-suction peak to double-suction peaks, and the pressure plateau region turns into a pressure collapse region. For the profiles of C_f (see Figs. 19(c)-(f)), four zero points can be identified in the transition region. Specifically, C_f first changes from positive to negative after the formation of primary vortex, and then reverts to positive again due to the generation of secondary vortex. After reattachment of the secondary separation vortex, C_f re-enters the primary vortex region, leading to another conversion from positive to negative values. Finally, with the reattachment of primary vortex, C_f shifts from negative to positive values again.

As a supplement to the aforementioned findings, quantities such as mean streamwise velocity and Reynolds shear stress are calculated as shown in Figs. 21 and 22. As aforementioned, no apparent difference exists among the SST- γ , Non-DaC, and DaC models for steady case of $AoA = 5^\circ$. In unsteady situation of $AoA = 17^\circ$, the SST- γ and DaC models are still consistent with each other, whereas a significant deviation is observed for the Non-DaC model. More critically, the qualitative difference outweighs the magnitude discrepancy. For the SST- γ and DaC models, two nontrivial zero points located in the secondary vortex region (i.e., $X/C = 0.0223$ in Fig. 21(b)) are observed on

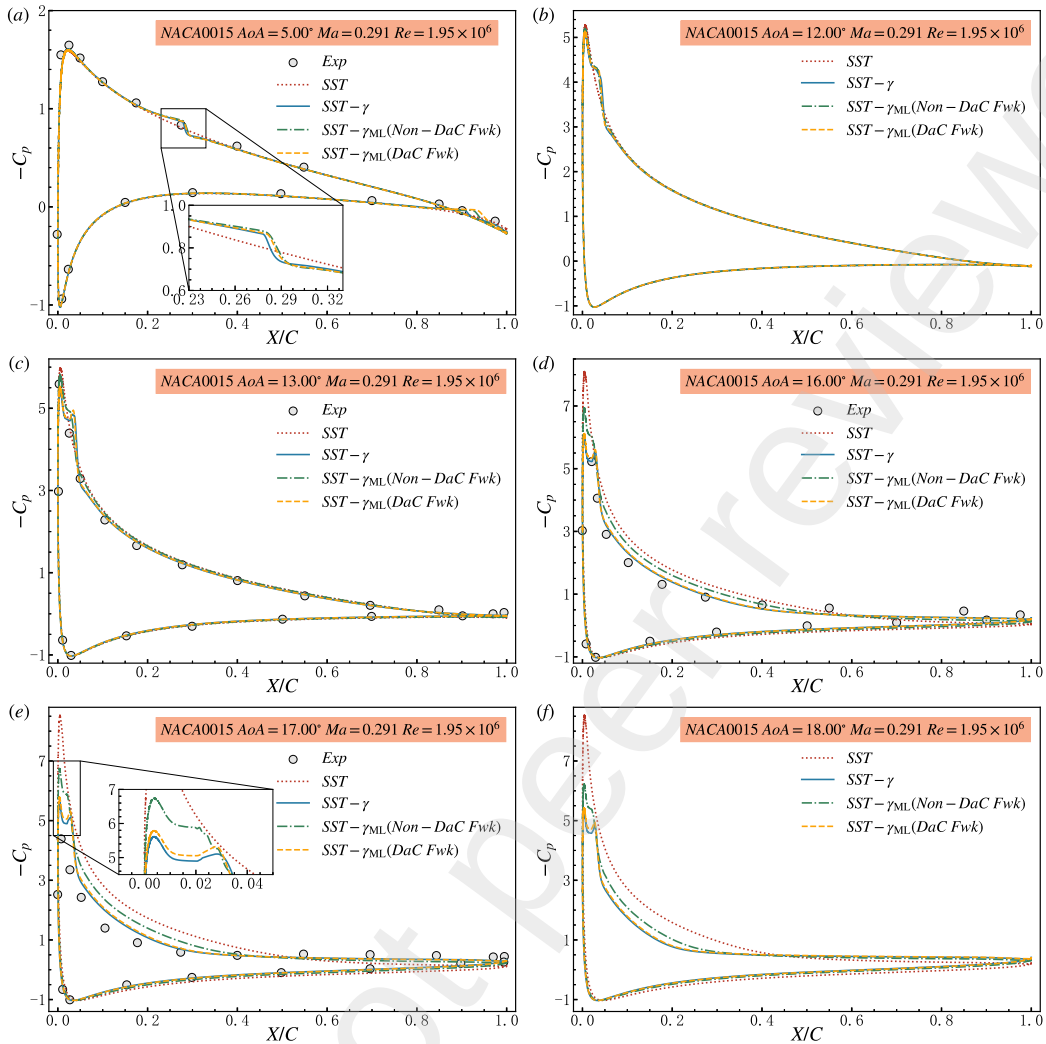


Fig. 18. Distributions of mean pressure coefficient C_p around the airfoil NACA0015 from present numerical simulations. The experimental data from Piziali [42] are plotted for comparison, and the corresponding flow regime is labeled in each panel for clarity.

the profile of mean streamwise velocity along the wall-normal direction owing to the secondary and primary vortex. Similar phenomenon is observed for the profile of Reynolds shear stress (see Fig. 22(b)). By contrast, the Non-DaC model predicts at most one nontrivial zero point because it can only capture a steady LSB with single vortex.

Overall, by using the divide-and-conquer strategy, substantial progress is made in the steady/unsteady integrated machine learning transition modeling. Compared with previous Non-DaC model in Sec. 2.2, the proposed DaC model achieves satisfactory solution to unsteady scenario, including the capture of stall characteristics, depiction of nonclassical LSB, and prediction of first/second-order velocity statistics, while maintaining high accuracy in simulation of flows in steady-state regime.

4.2. Generalization ability in AoA , Mach number, Reynolds number, and airfoil geometry

To facilitate the potential engineering application of present DaC-based SST- γ_{ML} transition-turbulence model, a comprehensive assessment is conducted for its generalization across AoA , Mach number, Reynolds numbers, and airfoil geometry. Firstly, the asymmetric NACA2414 airfoil is evaluated within the entire range of AoA s at $Ma = 0.250$ and $Re = 1.95 \times 10^6$. As depicted in Fig. 23, the DaC model achieves excellent agreement with the baseline SST- γ model from steady linear-lift stage to unsteady deep-stall stage. By contrast, the lift and drag coefficients are respectively

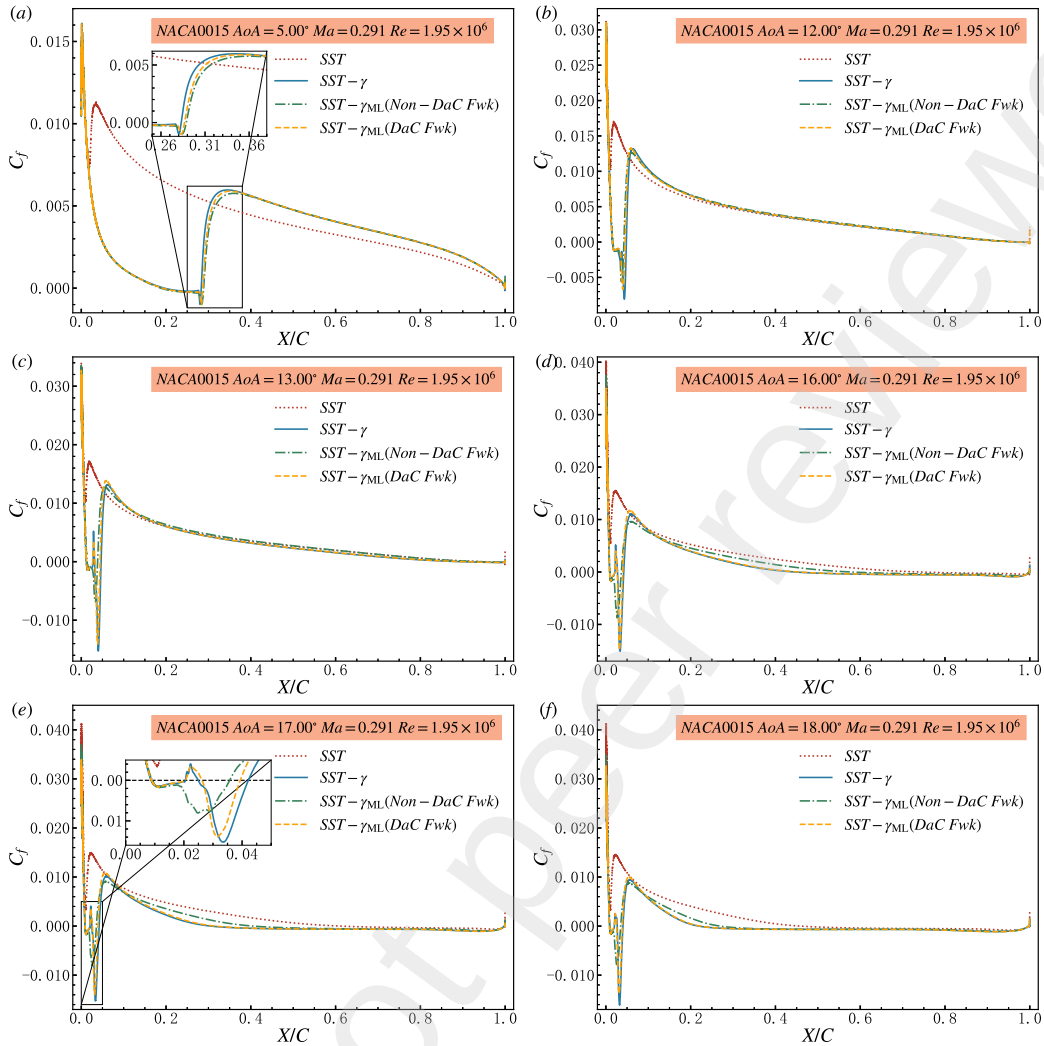


Fig. 19. Distributions of mean skin friction coefficient C_f around the airfoil NACA0015 (suction side only) from present numerical simulations. The experimental data from Piziali [42] are plotted for comparison, and the corresponding flow regime is labeled in each panel for clarity.

overestimated and underestimated by the original Non-DaC model, which contributes to a catastrophic misprediction of stall delay. Moreover, the fully turbulent SST model even demonstrates a more pronounced deviation than the Non-DaC model. Such performance also holds for the mean pressure coefficient C_p and mean skin friction coefficient C_f as shown in Figs. 24 and 25 except for the case with $AoA = 15^\circ$, in which the nonclassical LSB with a tiny secondary vortex emerges for the DaC model while its benchmark model still preserves the steady LSB structure. The origin of this phenomenon lies in the incorrect identification of an ISR by the IFISR classifier. Such misprediction is more likely to happen near the boundary between steady and unsteady flow regimes as the IFISR classifier is a purely data-driven model and may not perform well with small datasets. Nevertheless, the discrepancies in lift and drag coefficients caused by this potential misprediction are a conservative bias, differing markedly from the non-conservative errors typically observed for the Non-DaC model. In principle, designing a larger model for the IFISR classifier could enable precise identification of IFISR for each specific flow situation. However, the overarching business of this paper is not to develop a general-purpose DaC-based SST- γ_{ML} model, but to mine the potential superiority of DaC strategy over Non-DaC methods in complex task with multiple flow physics.

To further validate the generalization ability for airfoil geometry, simulations are conducted on the Aerospatiale A-airfoil with $Ma = 0.300$ and $Re = 3.00 \times 10^6$. In contrast to the previously discussed cases for NACA0015 and

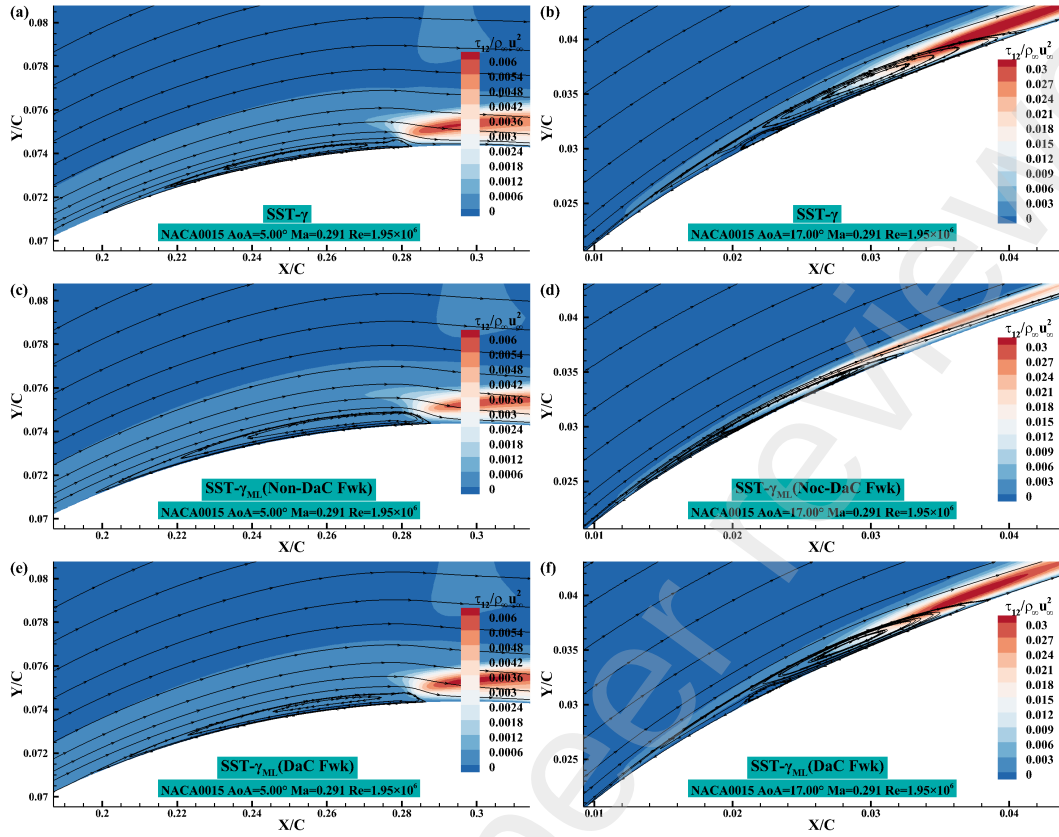


Fig. 20. Zoom-in streamlines around the transition region of the airfoil NACA0015 colored by Reynolds shear stress. For clarity, the employed turbulence models and corresponding flow regime is labeled in each panel.

NACA2414 airfoils, steady LSB spans through the entire range of AoAs (see Figs. 27 and 28). Therefore, computations can be completed solely via the steady-state time-marching LU-SGS method for all AoAs. Even so, the IFISR classifier of DaC model still accurately identifies the entire flow field as Non-ISR region, enabling the DaC model to fully align with Non-DaC model, and both exhibit excellent agreement with the baseline SST- γ model from linear-lift to stall stages, which demonstrates a broad generalization ability of the present DaC models across different airfoils and flow circumstances, despite the limited training cases.

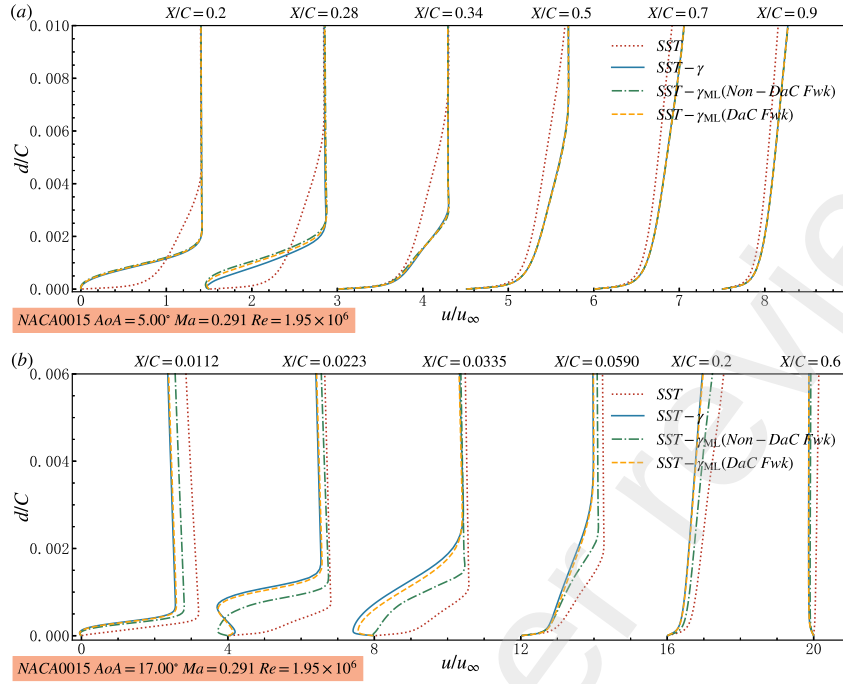


Fig. 21. Profiles of mean streamwise velocity along the wall-normal direction of NACA0015 airfoil with corresponding flow parameters labeled in each panel.

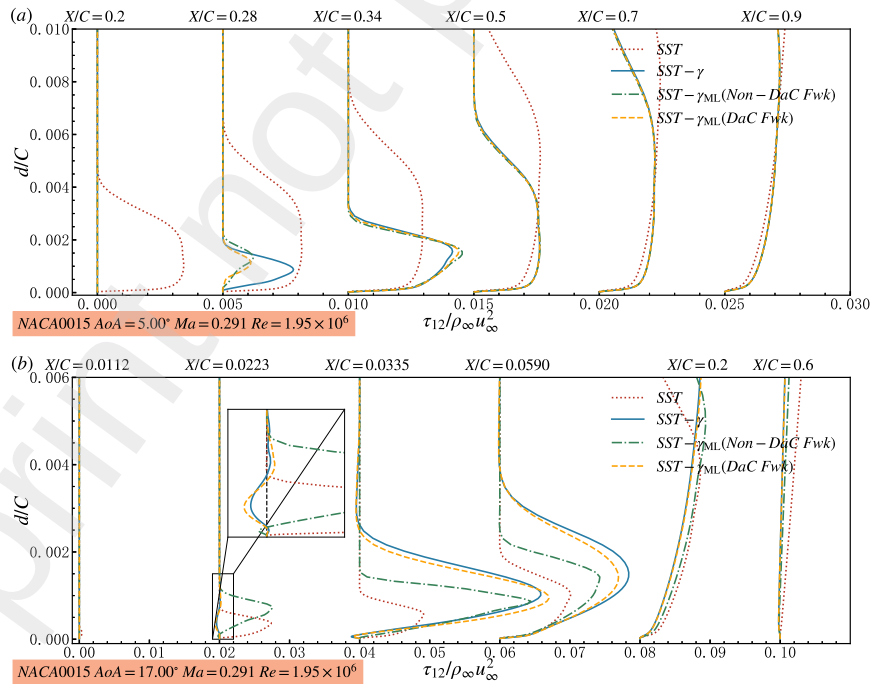


Fig. 22. Profiles of Reynolds shear stress along the wall-normal direction of NACA0015 airfoil with corresponding flow parameters labeled in each panel.

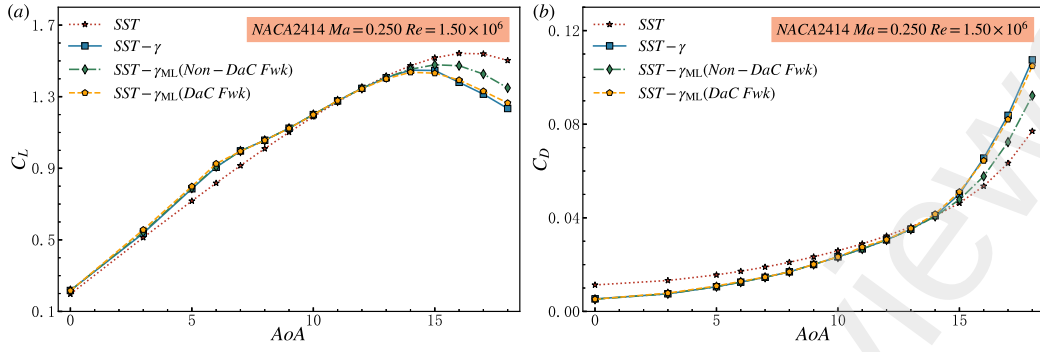


Fig. 23. (a) Lift coefficient and (b) drag coefficient versus angle of attack predicted by different RANS models for NACA2414 airfoil at $Ma = 0.250$ and $Re = 1.95 \times 10^6$.

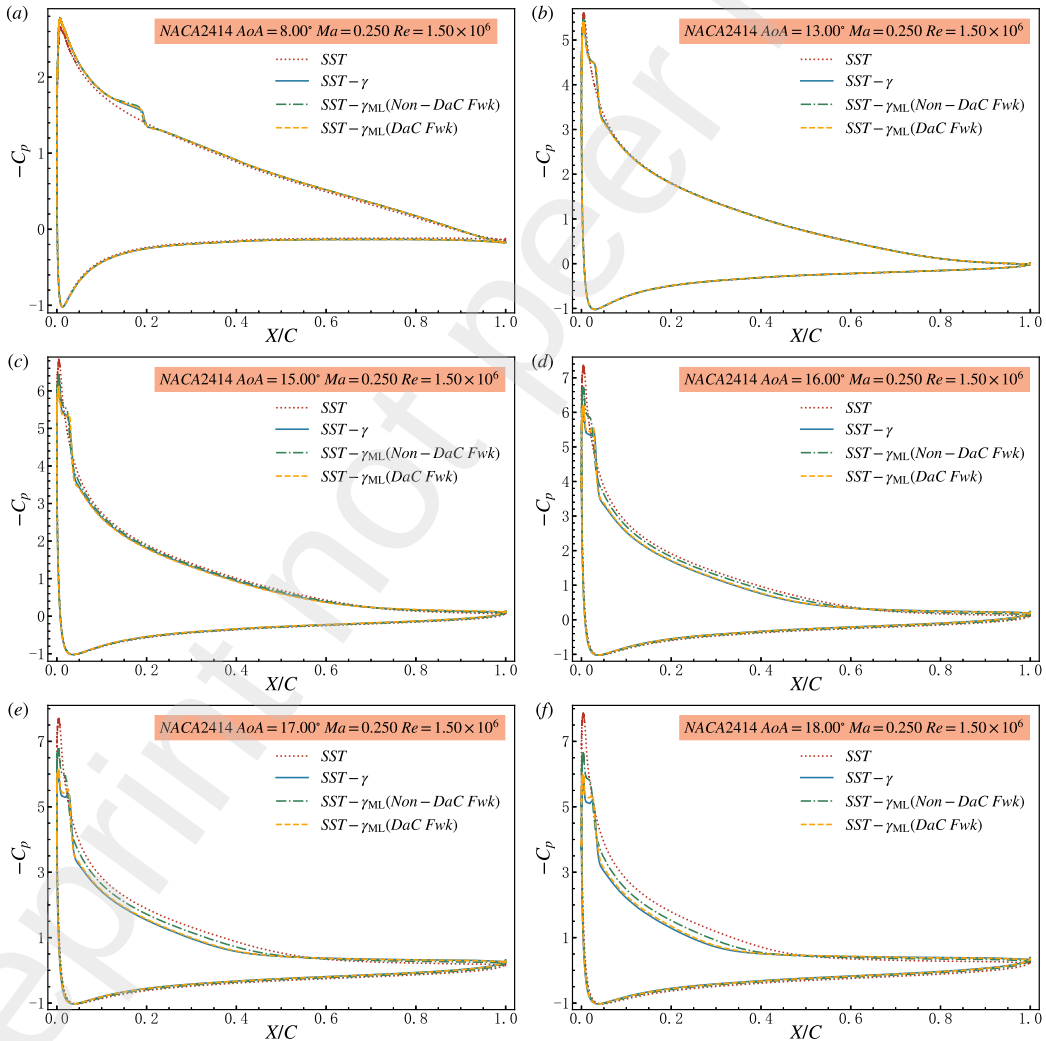


Fig. 24. Distributions of mean pressure coefficient C_p around the airfoil NACA2414 from present numerical simulations. Corresponding flow regime is labeled in each panel for clarity.

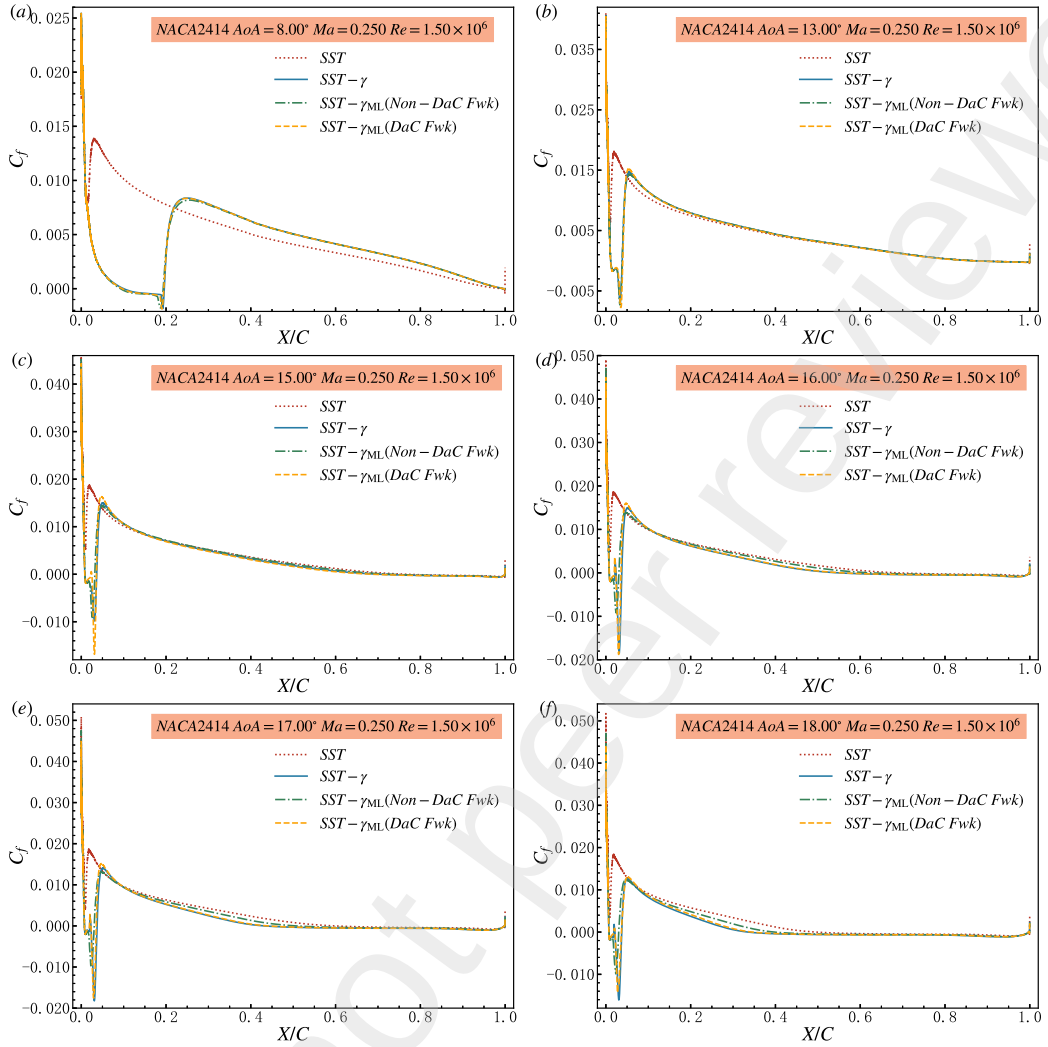


Fig. 25. Distributions of mean skin friction coefficient C_f around the airfoil NACA2414 (suction side only) from present numerical simulations. Corresponding flow regime is labeled in each panel for clarity.

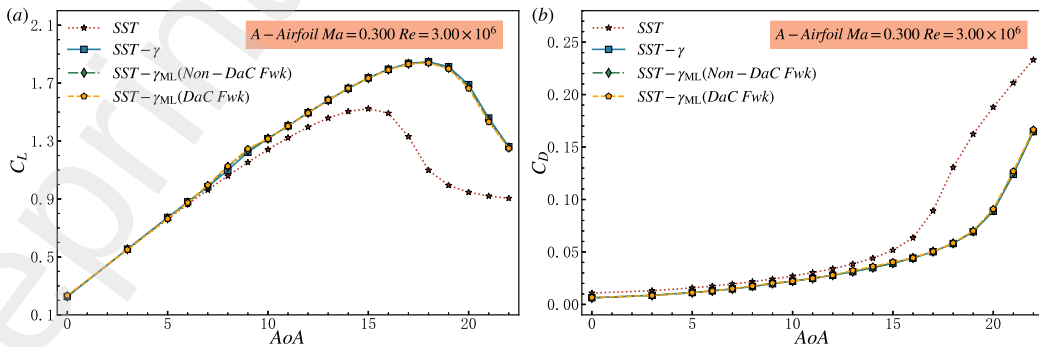


Fig. 26. (a) Lift coefficient and (b) drag coefficient versus angle of attack predicted by different RANS models for A-Airfoil at $Ma = 0.300$ and $Re = 3.0 \times 10^6$.

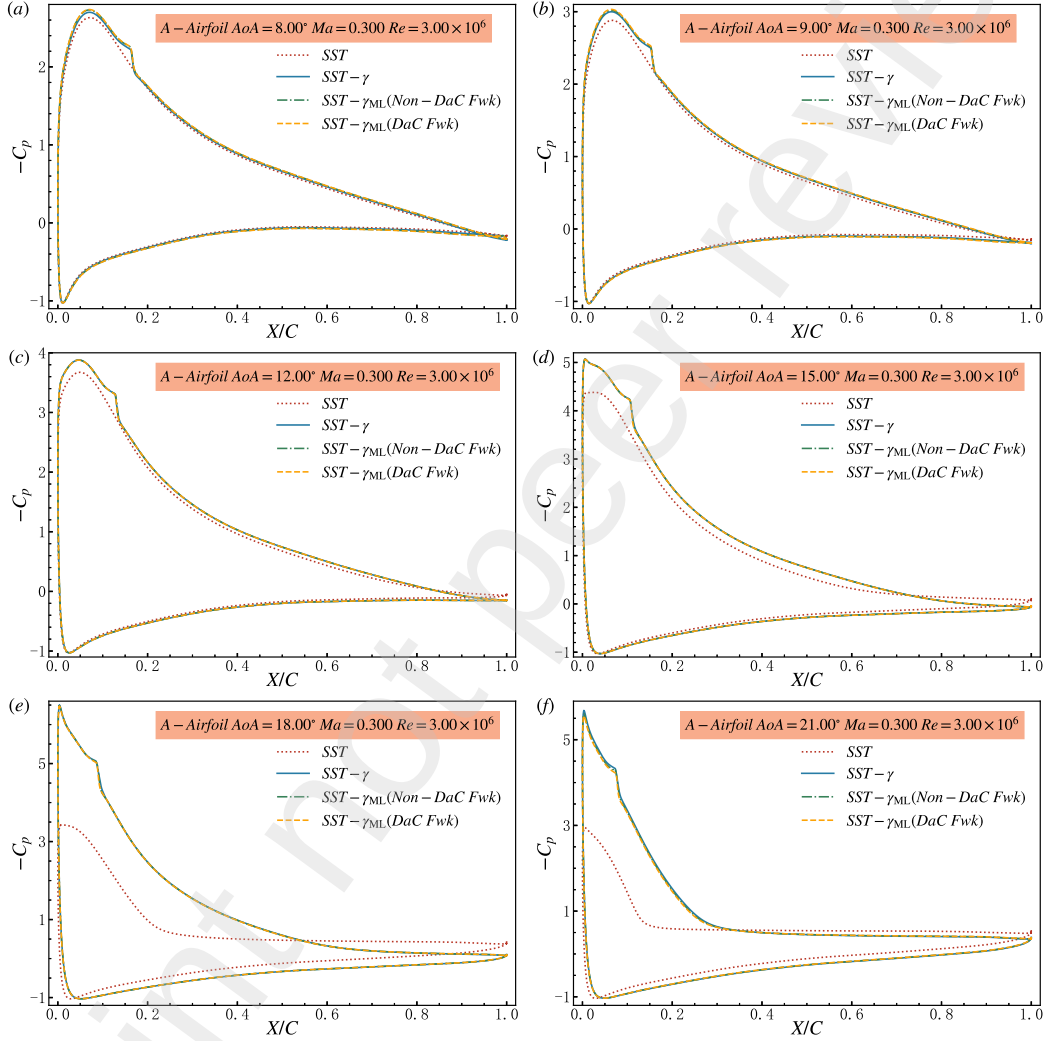


Fig. 27. Distributions of mean pressure coefficient C_p around the A-Airfoil from present numerical simulations. Corresponding flow regime is labeled in each panel for clarity.

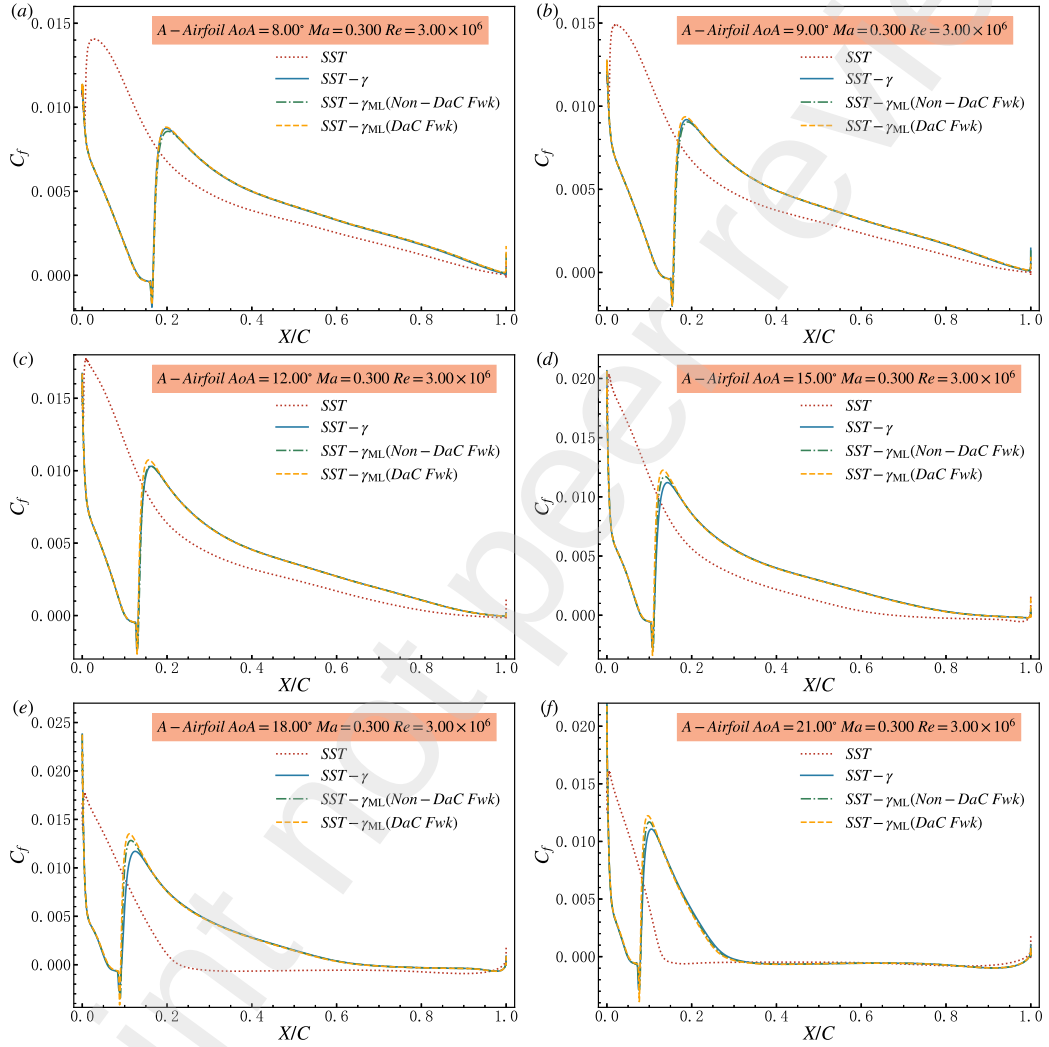


Fig. 28. Distributions of mean skin friction coefficient C_f around the A-Airfoil (suction side only) from present numerical simulations. Corresponding flow regime is labeled in each panel for clarity.

5. Conclusions and discussions

In this paper, a steady/unsteady integrated machine learning transition model is developed using the divide-and-conquer (DaC) strategy, which applies to a wide range from linear-lift to stall stages of airfoil flows. The universally-acknowledged multi-time fusion strategy is firstly employed to extend previous steady SST- γ_{ANN} model to unsteady scenarios. Preliminary *a priori* and *a posteriori* results reveal that the data fusion approach remains inadequate for addressing the present transition modeling issue. An in-depth inspection manifests that the core reason for the unsatisfactory performance of such model stems from the unresolved upstream intermittency spot region (ISR). In view of the intrinsic distinction of ISR from other region, a divide-and-conquer strategy is employed. A random forest classifier is constructed to determine whether the grid point falls within the ISR. For regression of the intermittency factor γ , artificial neural network and Gaussian process models are formulated to handle the Non-ISR and ISR, respectively. To verify the accuracy and generalization ability of the new DaC-based SST- γ_{ML} model, extensive *a posteriori* computations are conducted for various airfoils at different flow conditions. Overall, the DaC model exhibits a consistent predictive nature with the benchmark SST- γ model throughout the entire range of angles of attack, yielding a transformative improvement over the original Non-DaC model. Whole flow behaviors from linear-lift to stall stages of an airfoil are comprehensively described by DaC model. Therein, steady and unsteady regimes with qualitative discrepancies are simultaneously resolved, in contrast to the previous Non-DaC model which is exclusively confined to steady-state phenomena. Such steady/unsteady integrated machine learning transition model may serve as a more efficient yet equally accurate alternative to its traditional counterpart in overall evaluation of the aerodynamic performance of an airfoil.

In summary, by implementing the novel divide-and-conquer strategy, significant milestone is achieved in this study for machine learning-based transition modeling toward greater maturity and engineering practicability. However, this paper also raises several important questions that necessitate further investigation. First of all, considering the ease-of-use and computational efficiency, this study only partially models γ within ISR from the spatial input perspective. It would be valuable to explore an efficient and fully comprehensive regression model for γ in ISR. Secondly, despite the success of divide-and-conquer strategy in addressing the crux of the matter, pursuit of a Non-DaC yet fully CFD-compatible steady/unsteady integrated machine learning transition model still deserves attention, notwithstanding its potential complexity or brute-force nature. Finally, incorporating the data assimilation approach may be a generally adoptable strategy for present machine learning-based model to achieve higher prediction accuracy than the benchmark model.

Declaration of competing interest

The authors do not have any known competitive economic interests or personal relationships that may affect the work reported in this article.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Acknowledgment

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A. Candidate input features

A pool of candidate inputs are crafted to convey as much relevant local flow information as possible to the intermittency factor γ . These arguments are quantified and distilled into each individual ML-based model. For simplicity and completeness, we only sketch the basic principle of constructing candidate input features in this appendix. Readers may refer to other literature for more details [30, 34].

Firstly, the modeling of γ is highly coupled with the modeling of Reynolds stress. Wu et al. [58] have established a general functional mapping from mean flow quantities to Reynolds stress, $\tau^F = g(\mathbf{S}, \mathbf{\Omega}, \nabla p, \nabla k)$, with $\mathbf{S}$ and $\mathbf{\Omega}$

being the strain rate and rotation tensors, respectively. To generate minimal invariant bases from a set of symmetric and antisymmetric tensors, the gradients of pressure and TKE (∇p and ∇k) are transformed to the corresponding antisymmetric tensors $\mathbf{A}_p$ and $\mathbf{A}_k$. As a result, 17 minimal integrity bases consisting of all the traces of the independent matrix products are formed according to the Cayley-Hamilton theorem [59], which are detailed in Table A1. Secondly, considerable efforts are devoted to constructing input features with well-defined physical meanings [30, 34, 58, 60–64], which are appropriately chosen and summarized in Table A2. Each input makes specific contribution in its own way, such as amount of turbulence (f_6, f_9), identification of boundary layer ($f_1, f_2, f_3, f_4, f_{12}$), and indicator of flow property and structure (f_5, f_7, f_{11}). Finally, as a more straightforward way, incorporating the domain knowledge of conventional transition-turbulence model may contribute to the depiction of physical relevance between RANS mean flow variables and γ . In such endeavors, several functions are extracted as listed in Table A3. Readers are referred to the paper by Menter et al. [28] for their detailed formulations.

Table A1

Candidate input features q_1 - q_{17} from the Reynolds stress representation perspective. Here, $\hat{\mathbf{S}} = \frac{\mathbf{S}}{\|\mathbf{S}\| + \varepsilon/k}$, $\hat{\mathbf{\Omega}} = \frac{\mathbf{\Omega}}{2\|\mathbf{\Omega}\|}$, $\hat{\mathbf{A}}_p = \frac{\mathbf{A}_p}{\|\mathbf{A}_p\| + \rho|(\mathbf{u} \cdot \nabla) \mathbf{u}|}$, and $\hat{\mathbf{A}}_k = \frac{\mathbf{A}_k}{\|\mathbf{A}_k\| + \varepsilon/\sqrt{k}}$ with ε being the turbulent dissipation rate. The notations $|\cdot|$ and $\|\cdot\|$ represent the vector and Frobenius matrix norms, respectively.

Features q_i	Invariant bases
q_1	$\hat{\mathbf{S}}^2$
q_2 - q_3	$\hat{\mathbf{A}}_p^2, \hat{\mathbf{A}}_k^2$
q_4 - q_8	$\hat{\mathbf{\Omega}}^2 \hat{\mathbf{S}}^2, \hat{\mathbf{A}}_p^2 \hat{\mathbf{S}}, \hat{\mathbf{A}}_p^2 \hat{\mathbf{S}}^2, \hat{\mathbf{A}}_k^2 \hat{\mathbf{S}}, \hat{\mathbf{A}}_k^2 \hat{\mathbf{S}}^2$
q_9	$\hat{\mathbf{A}}_p \hat{\mathbf{A}}_k$
q_{10} - q_{13}	$\hat{\mathbf{A}}_p^2 \hat{\mathbf{\Omega}} \hat{\mathbf{S}}, \hat{\mathbf{A}}_k^2 \hat{\mathbf{\Omega}} \hat{\mathbf{S}}, \hat{\mathbf{A}}_p \hat{\mathbf{A}}_k \hat{\mathbf{S}}, \hat{\mathbf{A}}_p \hat{\mathbf{A}}_k \hat{\mathbf{S}}^2$
q_{14}	$\hat{\mathbf{\Omega}} \hat{\mathbf{A}}_p \hat{\mathbf{A}}_k$
q_{15} - q_{17}	$\hat{\mathbf{\Omega}} \hat{\mathbf{A}}_p \hat{\mathbf{A}}_k \hat{\mathbf{S}}, \hat{\mathbf{\Omega}} \hat{\mathbf{A}}_p \hat{\mathbf{A}}_k \hat{\mathbf{S}}^2, \hat{\mathbf{\Omega}} \hat{\mathbf{A}}_k \hat{\mathbf{A}}_p \hat{\mathbf{S}}^2$

Table A2

Candidate input features f_1 - f_{16} from other research works. Here, α and β refer to the raw feature and normalization factor, respectively. The normalization scheme $\alpha = \alpha/(|\alpha| + |\beta|)$ is executed for these inputs [65]. In addition, $\kappa = 0.41$ is the Karman constant and d is distance to the nearest wall.

Features f_i	Description	Formula
f_1	Streamwise velocity	u
f_2	Temperature	T
f_3	Boundary layer identification function	$1 - \tanh(8r_d)^3$ $r_d = \frac{\mu_T + \mu_L}{\kappa^2 \rho d^2 \sqrt{\partial_j u_i \partial_j u_i}}$
f_4	Modification of f_3	$1 - \tanh(\sqrt{r_d})$
f_5	Q-criterion	$\alpha = 0.5(\ \mathbf{\Omega}\ ^2 - \ \mathbf{S}\ ^2)$, $\beta = \ \mathbf{S}\ ^2$
f_6	Turbulence intensity	$\alpha = k$, $\beta = 0.5 \mathbf{u} $
f_7	Wall-distance-based Reynolds number	$\min(\frac{\sqrt{k} \rho d}{50 \mu_L}, 2)$
f_8	Pressure gradient along streamline	$\alpha = u_k \partial_k p$, $\beta = \sqrt{\partial_j p \partial_j p} u_i$
f_9	Eddy viscosity ratio	$\alpha = \mu_T$, $\beta = 100 \mu_L$
f_{10}	Ratio of pressure normal stresses to shear stresses	$\alpha = \sqrt{\partial_i p \partial_i p}$, $\beta = 0.5 \rho \partial_k u_k^2$
f_{11}	Non-orthogonality between velocity and its gradient	$\alpha = u_i u_j \partial_j u_i $, $\beta = \sqrt{u_i u_i \partial_j u_i \partial_j u_i}$
f_{12}	Wall distance function	$\exp^{-1}(d)$
f_{13}	Deflection angle of streamline	$A \circ A - \arctan(v/u)$
f_{14}	Refined entropy	$p/\rho^\gamma - 1$, $\gamma = 1.4$
f_{15}	Streamline curvature	$\frac{ D\Gamma }{D_s}$, $\Gamma = \frac{\mathbf{u}}{ \mathbf{u} }$, $D_s = \mathbf{u} Dt$
f_{16}	Tangential velocity relative to the wall	$u \sin(\theta) - v \cos(\theta)$

Table A3Candidate input features p_1 – p_{11} from traditional SST- γ model [28].

Features p_i	Description	Formula
p_1	Pure production of k equation	$\tau_{ij}^F S_{ij} - \beta^* \rho \omega k$
p_2	Pure production of ω equation	$\frac{C_{\omega\rho}}{\mu_T} \tau_{ij}^F S_{ij} - \beta \rho \omega^2$
p_3	Model function in $Re_{\theta c}$ (divided by ten)	$Tu_L/10$
p_4	Model function in $\lambda_{\theta L}$	dV/dy
p_5	Model function in F_{PG}	$\lambda_{\theta L}$
p_6	Model function in $Re_{\theta c}$	F_{PG}
p_7	Critical momentum thickness Reynolds number of γ equation (logarithmic form)	$\lg Re_{\theta c}$
p_8	Vorticity Reynolds number	Re_V
p_9	Model function in F_{turb}	R_T
p_{10}	Model function in destruction term of γ equation	F_{turb}
p_{11}	Model function in production term of γ equation	F_{onset}

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